

Kumar Utkarsh

Indian Institute of Technology Guwahati, India
Bachelor of Technology in Data Science and Artificial Intelligence

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2025 – Present
GPA: 7.94/10.0

PROJECTS

Gas Optimizer

Feb 2026 – Mar 2026

Kriti 2026, Inter Hostel Technical Competition, IIT Guwahati

[GitHub Project Link](#)

- Built a gasless transaction platform on Ethereum enabling users to execute batched smart contract calls without holding ETH, using EIP-712 typed signatures, EIP-2771 trusted forwarding, and a Fastify relayer service.
- Developed a React + Vite + wagmi frontend with real-time relayer health monitoring, wallet-based auth, and a dashboard for batch submission and gas analytics deployed on Vercel with a Railway-hosted relayer.
- Achieved ~45 percent gas savings on multi-call batches vs sequential transactions, validated through Foundry unit, fuzz, invariant, and integration tests across 7 core Solidity contracts on Sepolia testnet.

BDH Interpretability explorer

Jan 2026 – Feb 2026

Kriti 2026, Inter Hostel Technical Competition, Pathway

[GitHub](#)

- Built a 9-page interactive visualization platform using React, TypeScript, Three.js, and FastAPI for exploring a biologically-inspired post-transformer language model.
- Trained multilingual BDH models on Europarl corpus with PyTorch (bfloat16, A100) and merged via neuron concatenation into a 24,576-neuron polyglot model.
- Implemented monosemanticity probing, Hebbian synapse tracking, 3D graph topology, and sparsity analysis with a 6-phase extraction pipeline deployed on Hugging Face Spaces.

INITIA Gaming Infrastructure

Apr 2026

Hackathon Project, INITIATE Hackathon

[GitHub](#)

- Built a permissionless gaming infrastructure protocol on Initia MiniEVM where a single registerGameWithFee() call auto-deploys an ERC-20 token, ERC-1155 item collection, and AMM DEX pool; integrated ERC-6551 Token Bound Accounts for unified cross-game player identity across 20 deployed contracts.
- Engineered a GasPayermaster (ERC-2771 meta-transactions) enabling players to pay gas in any game token via an on-chain DEX swap to PXL, eliminating the need to hold native gas tokens; implemented IBC bridging from Solidity using Initia's ICosmos precompile (0xF1) for ERC-20 and ERC-1155 transfers to L1.
- Shipped a React + Vite frontend with @initia/interwovenkit-react session keys for zero-friction signing, a self-hosted MiniEVM node (VPS + nginx + systemd), and 1,545 lines of Hardhat tests across 12 files including a full E2E integration suite.

ACHIEVEMENTS

- Smart India Hackathon (SIH)** | Led the team in Blockchain PS, secured 4th position, and selected in the top 30 teams of the campus to qualify for Nationals. **2026**
- Codeforces** | 1344 Rated, profile - Kumar_Utkarsh9263 **2025**
- Octant v2 DeFi Hackathon** | Got the most Liked project and got their POAP token for free as reward. **2025**
- Codechef** | 3 Star(1613 rating) ranked 23 in Starters 234 contest, username - neat_glim_65 **2025**
- World Quant Brain Challenge** | Secured Team Rank of 2000 among 80,000+ teams **2025**

TECHNICAL SKILLS

Programming languages: C, C++, Solidity, Python
Web Technologies: HTML, CSS, ReactJS, Vercel, php
Database Management: MySQL, IPFS
Finance: Blockchain, Fundamental Analysis
Miscellaneous: Git, GitHub, Windows, Linux

RELEVANT COURSEWORK

Computer Science: Data Structures and Algorithms (Theory & Lab), Data Science (Theory & Lab), Discrete Mathematics
Mathematics: Probability Theory and Random Processes, Elementary Number Theory, Linear Algebra, Real Analysis and Vector Calculus

POSITIONS OF RESPONSIBILITY

De-Fi and Blockchain Head, Finance and Economics Club, IIT Guwahati

Apr 2026 – Present

- Leading cross-functional teams of **25+ developers** and executing strategic partnerships within the Web3 Ecosystem.